AYMANE AIT DADS

AI Software Development Intern | Agentic AI · End-to-End ML Pipelines · LLM Fine-Tuning

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PROFESSIONAL SUMMARY

Engineering student delivering production-grade AI pipelines, including an automated Random Forest + K-Means system on 100K+ fiber-optic records that reduced manual analysis time by 60% and directly informed network optimization decisions at Orange Maroc. Ranked top 10% on a $106K+ Kaggle competition leaderboard (0.80/1.0) fine-tuning a 30B Mamba-Transformer MoE model with LoRA on a Blackwell GPU, resolving 5 CUDA-level incompatibilities to enable training. Brings hands-on experience in AI agent integration, LLM orchestration, RAG, tool-calling, end-to-end software development, and cross-functional stakeholder delivery using Python, PyTorch, and Hugging Face Transformers. Available from September 2026 for a minimum of 20 weeks, authorized to intern in Europe.

CORE COMPETENCIES

AI Software Development | Agentic AI Platform | End-to-End Pipeline | LLM Orchestration | Smart Automation | Customer Engagement | Scalable Communication | Innovative Product Development

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built an end-to-end automated ML pipeline using Random Forest and K-Means on **100K+ fiber-optic network records**, classifying quality across thousands of nodes for the network optimization team.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, output adopted directly in production network planning decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation, and reproducible Scikit-learn workflows integrated into the team's reporting cycle.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, visualized via **Power BI** dashboards for non-technical decision-makers.

PROJECTS

NVIDIA Nemotron Reasoning Challenge *Kaggle Competition · In Progress June 2026 · Top 10% Public Leaderboard*

- Achieved **top 10% public leaderboard score (0.80/1.0) in a $106K+ prize-pool competition** by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) via SFTTrainer on 7,828 chain-of-thought logic examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** - including OOM monkey-patches, Mamba CUDA kernel stubs, and Triton ptxas fixes - and ran 20+ single-variable ablations confirming completion-only loss and packing=False as critical training decisions.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection *EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class*

- Ranked **1st in EURECOM class private Kaggle leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual learning rates, then submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Implemented a **3-model ensemble with 10-view test-time augmentation** across 250K training images and 25+ generator types, with FP16 mixed precision training on T4 GPU, finding that 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC) for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · Kaggle GPU · AdamW

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Web Applications, Image Security, Statistics

CPGE - Mathematics & Physics - Ibn Timiya *2021 - 2023*

SKILLS

ML Frameworks & AI Engineering: PyTorch | Hugging Face Transformers | Scikit-learn | Unsloth | TRL | PEFT | OpenCLIP

LLM & Agentic AI: LoRA | SFTTrainer | vLLM | LLM Orchestration | RAG | Tool-Calling | LangChain | Claude API

Software Development & Data: Python | SQL | Bash | Pandas | Git | Docker | PostgreSQL | Power BI